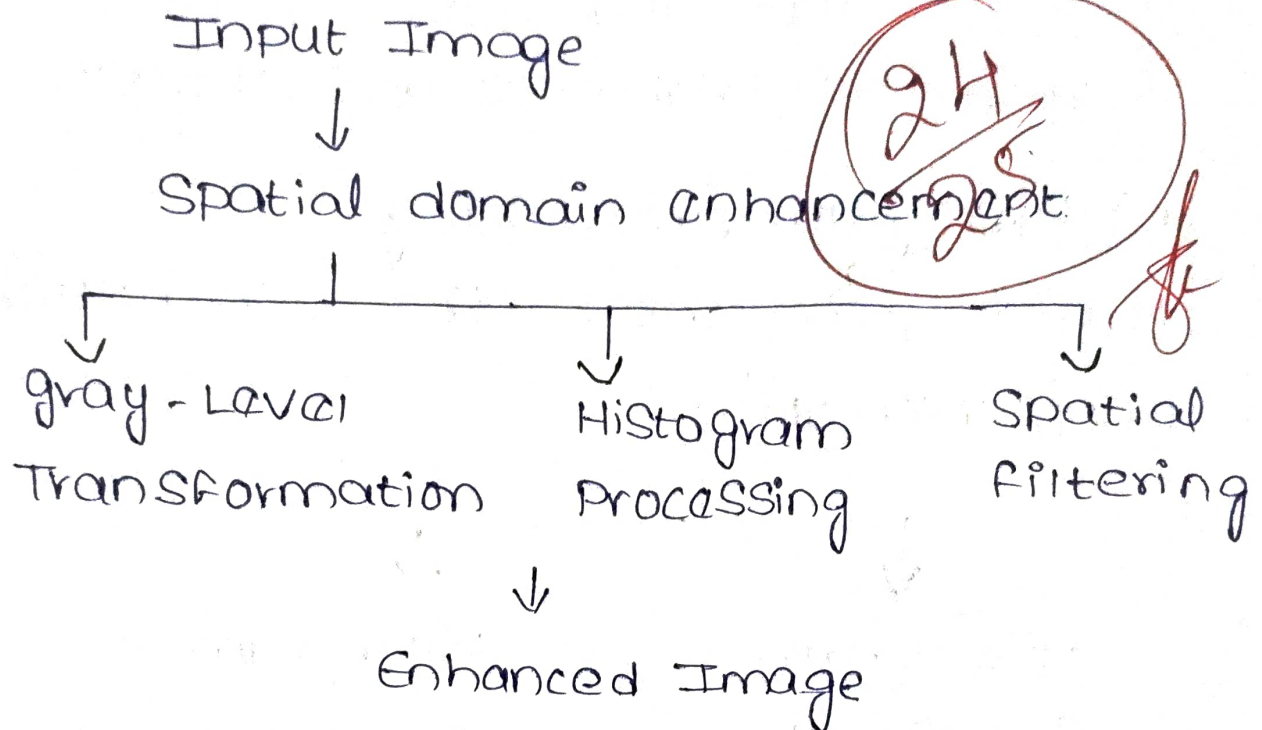


1. Image Enhancement In spatial domain



⇒ Image Enhancement In the spatial domain improves the quality of Image by directly modifying the pixel values.

⇒ It enhances brightness, contrast and important details while reducing noise

$$g(x, y) = T[f(x, y)]$$

↓ ↓ → Input Image

enhanced Image transformation function

(a) Gray Level Transformation: It modify the intensity values of pixels.

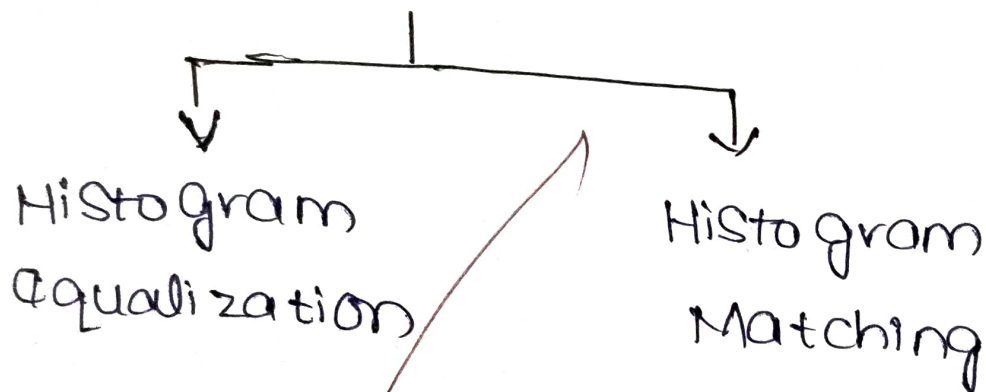
ex: Image negative
Log Transformation
Power-law Transformation

⇒ It is Simple and Fast

⇒ Improves brightness and contrast

(b) Histogram processing

A histogram shows the distribution of pixel intensities.



⇒ Improves overall contrast.

⇒ enhances dark or bright Images.

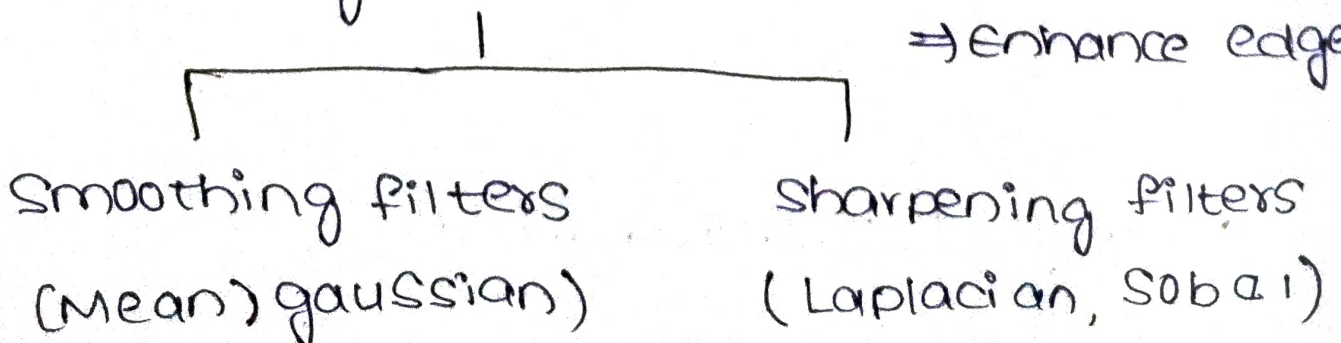
⇒ Suitable for poor lighting conditions, X-ray images

(c) Spatial filtering:

It uses a small matrix to process the neighboring pixels.

⇒ Remove noises

⇒ Enhance edges



Conclusion:

Gray level transformation improve brightness and contrast, Histogram processing enhances global contrast and spatial filtering removes noise or sharpens edges.

2. Image Smoothing vs Image Sharpening

→ Reduces noise

→ Removes fine details

→ uses low pass filters

→ Detect edges

→ Highlights fine details

→ uses high-pass filters

Spatial Smoothing Filters

→ These filters average neighboring pixel values

Ex: Mean filter

Median filter

Gaussian filter

⇒ Replace each pixel value with average or average of neighboring pixels

Advantages:

→ Removes noise

→ Smooths Image

→ Removes unwanted variations

Applications:

→ Medical Imaging

→ CCTV Image

→ preprocessing before Segmentation

(b) Sharpening Filters:

Sharpening filters emphasize rapid intensity changes

ex: Laplacial Filter

→ Sobel Filter

→ Prewitt Filter

working principle:

use Image derivatives to detect intensity changes and strengthen the edges.

Advantages:

→ Improves edges

→ Improves the visibility of the Image

Applications:

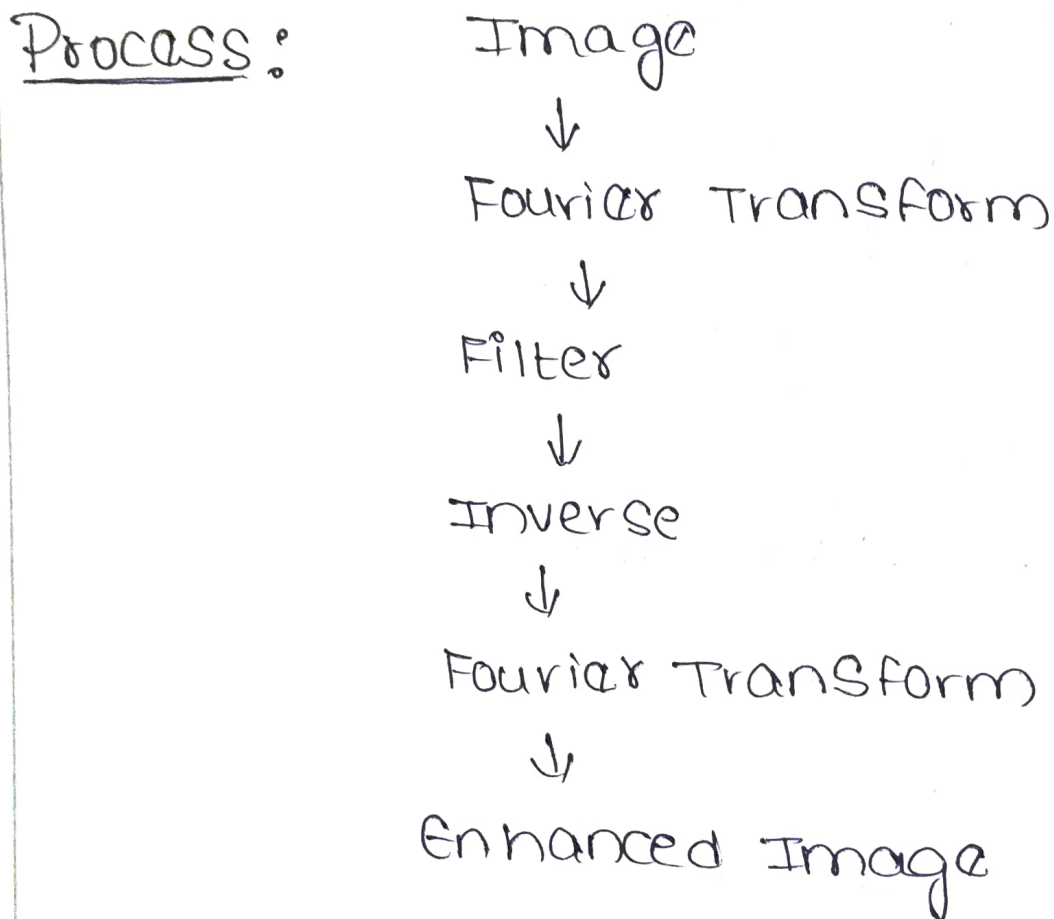
→ Fingerprint Recognition.

→ Industrial Inspection.

→ Face Recognition.

3. Image Enhancement in Frequency Domain

In the Frequency domain, Enhancement is performed after converting the image using the Fourier Transform



Low Pass Filter :

Allows Low Frequencies and blocks high Frequencies.

High Pass Filter :

Allows high Frequencies and blocks Low Frequencies

Low Pass Filter affects:

Removes noise

Smooths image

Reduces Sharpness.

Advantages:

Good noise reduction

Smooth appearance.

Applications:

Medical Images

Satellite Images.

High Pass Filter affects

Enhance edges.

~~Sharpens image.~~

~~Highlights details.~~

Advantages:

→ Face recognition.

→ Industrial Inspection.

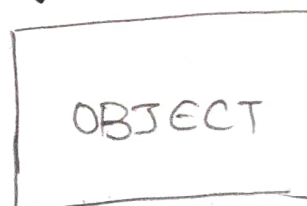
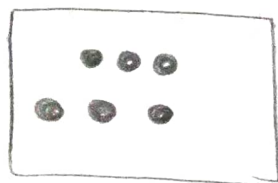
Conclusion:

LPF is preferred for noise reduction,
whereas HPF is suitable for edge
enhancement.

1. Image Segmentation

Image segmentation divides an Image into meaningful regions or objects for analysis.

Original Image $\rightarrow$ Segmented Image



Comparison :

Technique	working Principle	Advantage	Limitation
Point detection	detects isolated pixels	Simple	detects only single point
Line detection	Detects horizontal vertical diagonal lines	Identifies Line Structures	Sensitive to noise
Edge detection	Detects Intensity changes	Accurate boundaries	Noise Sensitive
Thresholding	Separate foreground background	Fast	Poor under uneven lighting

Region Based Segmentation	groups Similar neighbouring pixels	produce complete region	exp
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Most Appropriate Technique

For general Object detection:

Edge detection + Region Based detection

Justification:

- Edge detection accurately identifies Object boundaries
- Region based segmentation groups similar pixels into meaningful objects.

5. Edge detection, Edge Linking and Boundary detection

Original Image



Edge detection
(Detect edges)



Edge Linking
(Join broken edges)

↓
Boundary detection
(complete object outline)

↓
Object Recognition

Original boundary Edge Linked



(a) Edge detection :

detects sudden intensity changes that
Indicate object boundaries

Methods

→ Identifies object boundaries → Sobel

→ Prewitt

→ Feature extraction

→ Pattern Recognition

(b) Edge Linking :

connects broken edge pixels to form
continuous edges

⇒ Forms complete object shapes.

- Improves Segmentation
- Better object recognition.

(c) Boundary detection:

Extracts the complete outer boundary of an object.

→ Measures area and perimeter

→ Shape analysis

→ object classification.

Contribution to Object Recognition :

Edge detection : Finds important object features

Edge linking : Connect broken edges into continuous shapes

Boundary detection : Extracts complete object outlines for recognition and measurement.

ex : Medical Imaging

→ Detect tumor edges

→ Links boundary pixels

→ Extract complete tumor boundary.